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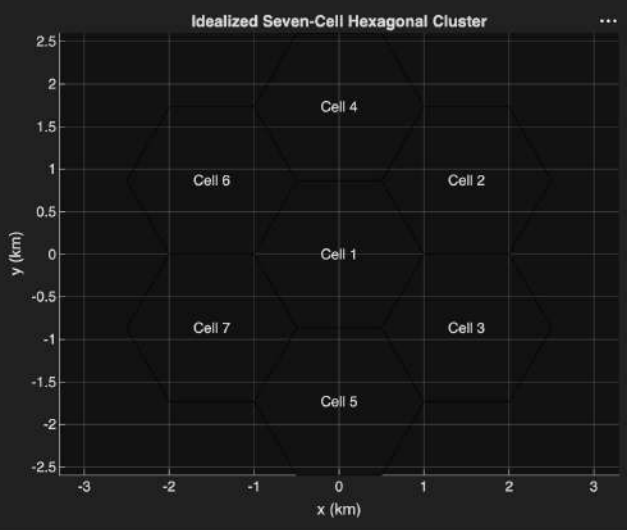
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```
% Baseline assumptions
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;
cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table(radiiKm(:), cellAreaKm2(:), cellCount(:), ...
    repmat(usersPerCell,numel(radiiKm),1), estimatedUsers(:), ...
    'VariableNames', {'Radius_km','CellArea_km2','EstimatedCells', ...
    'UsersPerCell','IllustrativeUsers'});
disp(capacityTable);
% Plot a 7-cell cluster: central cell plus six neighbors
R = radiiKm(1);
centers = [0 0; 1.5*R sqrt(3)*R/2; 1.5*R -sqrt(3)*R/2; ...
    0 sqrt(3)*R; 0 -sqrt(3)*R; -1.5*R sqrt(3)*R/2; ...
    -1.5*R -sqrt(3)*R/2];
theta = (0:6)*pi/3; % flat-top hexagon vertices
figure; hold on; axis equal; grid on;
for k = 1:size(centers,1)
    plot(centers(k,1)+R*cos(theta), centers(k,2)+R*sin(theta), 'k-');
    text(centers(k,1),centers(k,2),sprintf('Cell %d',k), ...
        'HorizontalAlignment','center');
end
xlabel('x (km)'); ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');

hold off;
% Mobile moves across the boundary between centers at x=0 and x=1.5R
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));
d1 = hypot(mobileX, mobileY);
d2 = hypot(mobileX-1.5*R, mobileY);
servingCell = ones(size(mobileX));
servingCell(d2 < d1) = 2;
handoffIndex = find(diff(servingCell) ~= 0,1,'first') + 1;
```

Figures

Figure 2 | Figure 3 | Figure 4




Command Window

```
    "5G"        "5G NR"    "OFDMA-based"    "Flexible OFDM-based radio access"

>> exp1cg
>> exp2g
    Radius_km    CellArea_km2    EstimatedCells    UsersPerCell    IllustrativeUsers
    _________    ____________    ______________    ____________    _________________

        1           2.5981             2               30                60
      0.5          0.64952             8               30               240

Geometric handoff at x = 0.753 km
>>
```

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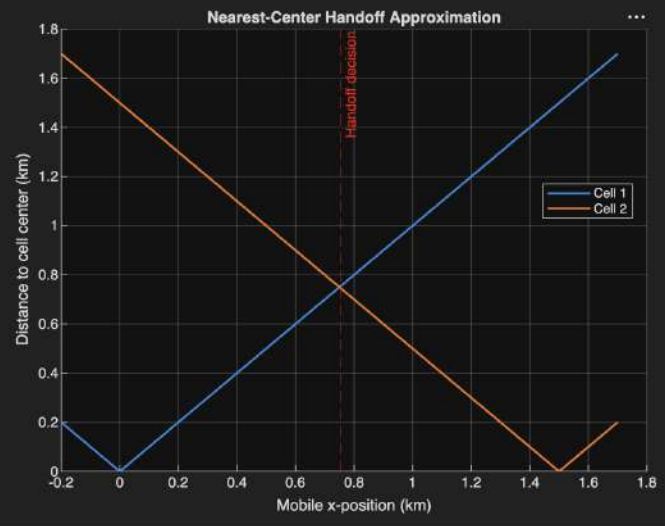
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Figures

Figure 2 | Figure 3 | Figure 4




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Geometric handoff at x = 0.753 km
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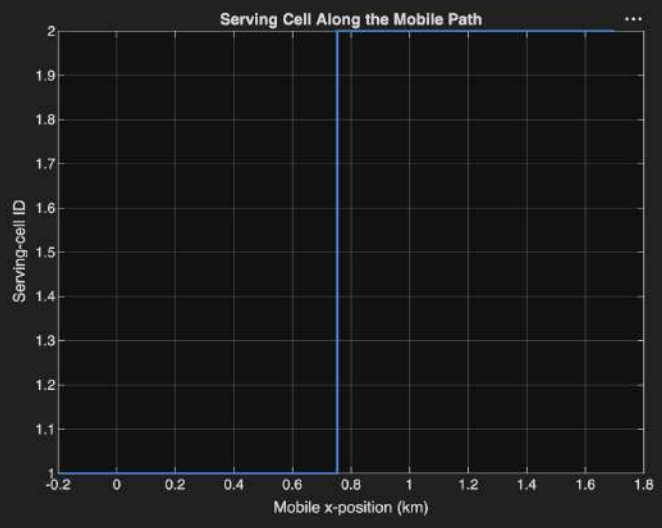
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**Figures**

Figure 2 | Figure 3 | Figure 4



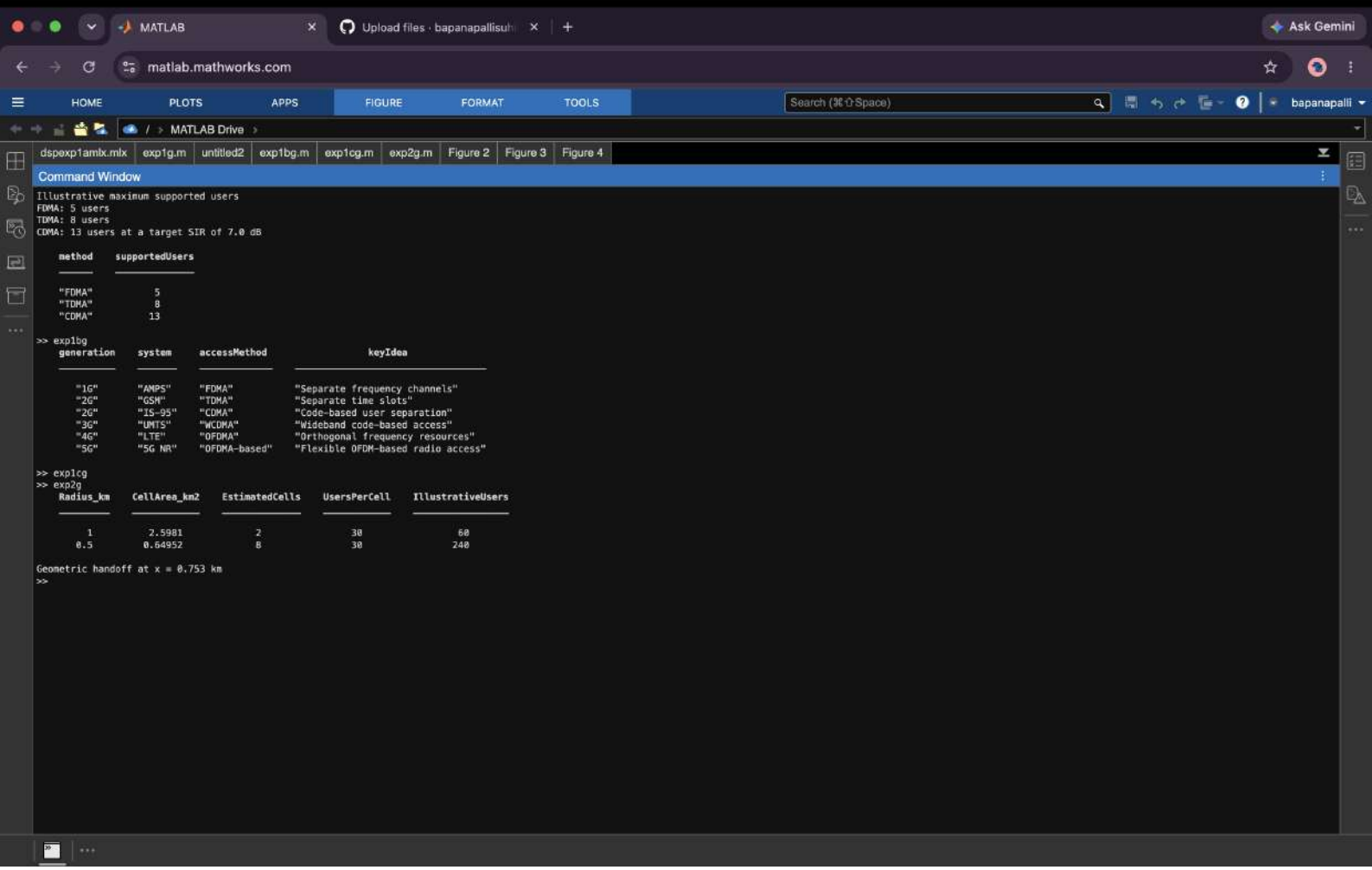

**Command Window**

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untitled2
exp1bg.m
exp1cg.m
exp2g.m
Figure 2
Figure 3
Figure 4
Command Window
Illustrative maximum supported users
FDMA: 5 users
TDMA: 8 users
CDMA: 13 users at a target SIR of 7.0 dB
method supportedUsers
"FDMA" 5
"TDMA" 8
"CDMA" 13
>> exp1bg
generation system accessMethod keyIdea
"1G" "AMPS" "FDMA" "Separate frequency channels"
"2G" "GSM" "TDMA" "Separate time slots"
"2G" "IS-95" "CDMA" "Code-based user separation"
"3G" "UMTS" "WCDMA" "Wideband code-based access"
"4G" "LTE" "OFDMA" "Orthogonal frequency resources"
"5G" "5G NR" "OFDMA-based" "Flexible OFDM-based radio access"
>> exp1cg
>> exp2g
Radius_km CellArea_km2 EstimatedCells UsersPerCell IllustrativeUsers
1 2.5981 2 30 60
0.5 0.64952 8 30 240
Geometric handoff at x = 0.753 km
>>